

1 Introduction

Fake news spreads quickly online and can mislead readers, and checking every article by hand is slow. This project builds an automatic **Fake News Detector** that reads a news headline or article and predicts whether it is **Fake** or **Real**, together with a confidence score. The trained model is wrapped in a simple web application so that anyone can use it without writing code.

2 Abstract

The system is trained on the Kaggle *Fake and Real News* dataset, which contains **39,105 unique articles** after duplicate removal (21,197 real and 17,908 fake). Text is cleaned and converted into numeric features using **TF-IDF**, then classified with **Logistic Regression**. On a held-out test set of 7,821 articles, the model achieved **98.25% accuracy**. A Streamlit app supports single-headline prediction and batch prediction through CSV upload, with downloadable results.

98.25%

Accuracy

0.98

Avg Precision

0.98

Avg Recall

0.98

Avg F1-score

3 Tools Used

Python 3.12	Programming language
VS Code + venv	Code editor and isolated virtual environment
pandas	Loading, merging and cleaning data; CSV handling
scikit-learn	TfidfVectorizer, Logistic Regression, train/test split, metrics
joblib	Saving and loading the trained model (.pkl files)
Streamlit	Interactive web interface (single and batch prediction)

4 Steps Involved in Building the Project



- Setup:** created the project in VS Code, set up a virtual environment, and installed pandas, scikit-learn, streamlit and joblib.
- Data loading:** loaded Fake.csv and True.csv, labelled them (0 = fake, 1 = real), merged them, removed duplicates and empty rows, and shuffled the data.
- Text cleaning:** combined title and body, converted to lowercase, and removed URLs, punctuation and numbers. The "(Reuters)" tag was also removed so the model does not learn from the agency name.
- Train/test split:** 80% training and 20% testing, stratified to keep the fake/real ratio equal.
- Feature extraction:** TF-IDF with English stop words removed, max_df = 0.7 and up to 50,000 features.
- Training:** fitted a Logistic Regression classifier (max_iter = 1000).
- Evaluation:** measured accuracy, precision, recall, F1-score and the confusion matrix on unseen test data.
- Saving:** stored the model and vectorizer as model.pkl and vectorizer.pkl.
- Web app:** built a Streamlit app with a text box (prediction and confidence) and a CSV uploader with a download button.
- Testing:** verified the app on sample real and fake headlines and a 3-row test CSV; all predictions were correct.

5 Results and Evaluation

The model was evaluated on 7,821 articles that it never saw during training. Both classes are detected with high precision and recall, and only 137 of 7,821 articles (1.75%) were misclassified.

Test-set results (7,821 articles)

Class	Precision	Recall	F1-score	Support
Fake	0.99	0.97	0.98	3,582
Real	0.98	0.99	0.98	4,239
Overall	0.98	0.98	0.98	7,821

Confusion matrix

	Predicted Fake	Predicted Real
Actual Fake	3,487	95
Actual Real	42	4,197

Sample headline (tested in the app)	Prediction	Confidence
Trump says he will sign the tax bill after Senate vote	Real	82.8%
SHOCKING: Hillary Clinton secretly arrested, media hiding the truth!!!	Fake	96.7%
Senate passes budget bill after lengthy debate	Real	79.1%

6 Conclusion

The project shows that TF-IDF with Logistic Regression is a fast and effective baseline for fake news detection, reaching **98.25% accuracy** with balanced precision and recall. The Streamlit app makes the model easy to use for single headlines and for bulk files. **Limitation:** the dataset mainly contains US political news from 2016-2017, so performance may drop on other topics or newer writing styles. **Future work:** train on more varied and recent data, try SVM or transformer models such as BERT, and deploy the app on Streamlit Cloud.